

209 Final Report

Introduction: Coronary Artery Disease (CAD) is a type of heart disease that occurs when there is too much plaque buildup in the arteries, leading to the heart not being able to deliver enough oxygenated blood to the rest of the body. About 1 in 20 adults over 20 have CAD, making it the most common type of heart disease; In the US in 2022 alone, it has caused 371,506 deaths. CAD falls under three categories: obstructive CAD (in which 50% or more of a large artery is blocked), non obstructive CAD (in which less than 50% of a large artery is blocked), and coronary microvascular disease (in which the tiny arteries in the heart are blocked) [1]. The risk factors for CAD include high bad cholesterol (LDL), low good cholesterol (HDL), high blood pressure, family history, diabetes, smoking, and obesity [2]. In general, CAD is preventable with a good diet and exercise, and if patients take medication, it's entirely possible to stop or even reverse the effects of plaque buildup [1]. The goal of this project is to figure out which clinical measurements can determine whether a patient has CAD.

Study Design: The data was downloaded from the UCI data repository, featuring datasets from Cleveland, Hungary, Switzerland, and VA Long Beach [3]. The Cleveland dataset consists of 303 patients, with a mean age of 54, who were referred for coronary angiography between 1981 and 1984. None of the patients had any prior evidence of heart attacks or cardiomyopathic disease. Their history was taken, along with a physical exam, ecg, serum cholesterol test, and fasting blood sugar. After consent was given, they also got an exercise ecg, thallium scintigraphy, and cardiac fluoroscopy. 200 patients were studied at the Veterans Administration Medical Center in Long Beach, 294 patients at the Hungarian Institute of Cardiology, and 123 patients across two university hospitals in Switzerland. For these three datasets, noninvasive tests and coronary angiography were taken for the patients. Histories and physical exams were given within 6 weeks before the coronary angiogram was given. If there was a greater than 50% narrowing of a major vessel, then the results for the coronary angiogram was considered abnormal [4]. All the datasets contain the columns age, sex (0: female, 1: male), chest pain (1: typical angina, 2: atypical angina, 3: non-anginal pain, 4: asymptomatic), resting blood pressure (resttbps), serum cholesterol (chol), fasting blood sugar (fbs), resting ecg (restecg with 0: normal, 1: ST-T wave abnormality, 2: left ventricle hypertrophy), maximum heart rate (thalach), exercise induced angina (exang, with 0: no, 1: yes), ST depression induced by exercise (oldpeak), slope of the peak exercise ST segment (slope, 1: upsloping, 2: flat, 3: downsloping), number of major vessels, 0 to 3, colored by fluoroscopy (ca), thallium scintigraphy results (thal, 3: normal, 6: fixed defect, 7: reversible defect), and diagnosis of heart disease (num, 0: <50% diameter narrowing, 1: >50% diameter narrowing) [3].

Data Analysis: Which of the variables listed above can predict whether a patient has CAD?

The first step was to pool the four datasets (Cleveland, Hungary, Switzerland, VA) into one dataset and add a column for the site. This way the dataset increases in size to 920 samples, and the results can be generalized beyond one hospital. The column num was then changed to target, and its coding became 0 for no CAD and 1 into any CAD. Then I checked whether any of the datasets had missing values in it. The following were the results.

site	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	num	target
<fct>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>
1 cleveland	0	0	0	0	0	0	0	0	0	0	0	4	2	0	0
2 hungary	0	0	0	1	23	8	1	1	1	0	190	291	266	0	0
3 switzerland	0	0	0	2	0	75	1	1	1	6	17	118	52	0	0
4 va	0	0	0	56	7	7	0	53	53	56	102	198	166	0	0

Figure 1 - Missing values by site

This is presumably due to the non Cleveland sites not collecting each variable data consistently. To fix this I chose to use multiple imputation over removing the null values, as removing those values would reduce the size of the dataset significantly, and would lead to the majority of samples coming from Cleveland. Since the missing data seems dependent on the site, it is considered Missing At Random, and thus a good fit for multiple imputation.

I used the mice package in R to create 20 versions of the dataset, with each version's missing values filled in a little differently using predictive mean matching. This method uses a regression (different types for different variables) to fill in missing values, based on the values of patients with similar observed values. I set mice to do this process 20 times, creating 20 versions of an imputed dataset. Below is a graph for each variable and how well the imputation worked.

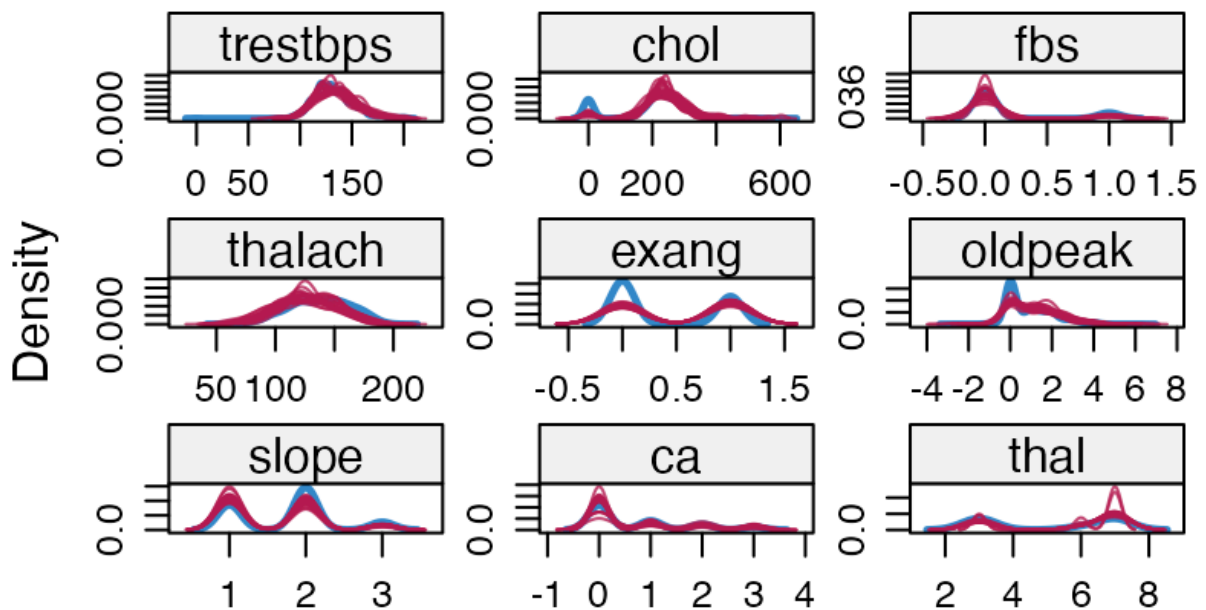


Figure 2 - Imputations: Blue represents the distribution of the observed values, and red represents the distribution of the imputed values in each of the 20 versions of the dataset. We want to see red and blue overlap, which means that the imputation did a good job, which for the most part we see. We do see discrepancies in chol, exang, and oldpeak, which is to be expected as those were the variables with the most missing values.

From there, I ran a logistic regression (glmer to account for the site variable, as each site has a differing baseline CAD rate due to diet, culture, and other variables, and to predict a binary outcome) on the 20 imputed datasets, resulting in 20 fitted glmer objects. I then used the pool function to pool the set of 20 results into one final result, using Rubin's rules. This averages the coefficient values across all 20 results into one value. But it works a little differently for the standard errors, because we technically guessed the values of the missing number. To fix this, it combines the uncertainty of each individual model with the variation between each model to get one standard error that's larger than what an ordinary model would give, but more accurate. Our results now are one coefficient (log odds) per variable, one standard error, and one p-value.

	term	estimate	std.error	statistic	df	p.value
1	(Intercept)	-5.7155233588	1.911083632	-2.9907238	175.46655	3.183499e-03
2	age	-0.0119992812	0.016309418	-0.7357271	195.24458	4.627797e-01
3	sex	0.8290661688	0.312094458	2.6564591	473.65135	8.163085e-03
4	cp	0.5966405685	0.131189704	4.5479222	273.90750	8.144149e-06
5	trestbps	0.0075504794	0.006566659	1.1498205	232.22590	2.514006e-01
6	chol	0.0008607288	0.001628371	0.5285829	241.47991	5.975803e-01
7	fbs	-0.0352878205	0.372652235	-0.0946937	140.12615	9.246934e-01
8	restecg	0.0760628101	0.146527645	0.5191021	709.86861	6.038513e-01
9	thalach	-0.0092631468	0.006056788	-1.5293827	185.29818	1.278749e-01
10	exang	1.1307681010	0.280765560	4.0274459	199.65784	8.008273e-05
11	oldpeak	0.4063280954	0.135494961	2.9988429	331.81711	2.914846e-03
12	slope	0.5648799612	0.250300298	2.2568090	113.40892	2.593577e-02
13	ca	1.3061263634	0.268673204	4.8613942	54.74200	1.020680e-05
14	thal	0.3717607838	0.090066204	4.1276391	66.55051	1.042575e-04

Figure 3 - Results

Sex, cp, exang, oldpeak, slope, ca, and thal are statistically significant. Males have 2.29 times the odds of CAD that females have. Each unit increase in chest pain is associated with 82% increase in the odds of CAD. Oldpeak and slope are associated with a 50% and 76% higher odds respectively. The two variables with the largest odds ratios are exang and ca. Each unit increase of exercise induced angina is associated with a 3.1 times higher odds of CAD, and each additional blockage is associated with a 3.69 times higher odds of CAD. Age, trestbps, chol, fbs, restecg, and thalach are not statistically significant. One interesting part about that is that cholesterol and blood pressure are considered to be risk factors for CAD, but once the model accounts for the stress test results (a far better predictor), they lose their predictive power. Overall, the model shows that the results of a stress test are far more predictive of CAD than any resting measurement.

I then ran a comparison model, removing the site random intercept, to check whether there is a statistically significant difference between adding it and not. The model will now treat all the patients as if they were from the same region. The results are below.

	term	estimate	std.error	statistic	df	p.value
1	(Intercept)	-5.685769086	1.914986740	-2.96909058	94.15682	3.788303e-03
2	age	-0.006932234	0.019146863	-0.36205587	53.00979	7.187496e-01
3	sex	0.838446673	0.320240162	2.61818089	235.91111	9.412170e-03
4	cp	0.632480732	0.133296587	4.74491319	196.16567	4.011164e-06
5	trestbps	0.007887986	0.006689394	1.17917793	146.43985	2.402396e-01
6	chol	-0.002365229	0.001827118	-1.29451405	36.56059	2.036046e-01
7	fbs	0.010221489	0.409366892	0.02496902	69.54373	9.801512e-01
8	restecg	0.129598987	0.150726671	0.85982784	243.63405	3.907292e-01
9	thalach	-0.009427458	0.005877354	-1.60403087	116.21123	1.114216e-01
10	exang	1.130331124	0.277887884	4.06757973	189.50837	6.962848e-05
11	oldpeak	0.375376180	0.129882976	2.89011071	308.68951	4.123938e-03
12	slope	0.613738814	0.251135692	2.44385340	91.27263	1.645390e-02
13	ca	1.386215233	0.288544903	4.80415775	38.04041	2.437864e-05
14	thal	0.331583601	0.086292723	3.84254417	62.18367	2.884558e-04

Figure 4 - Results of the Comparison Model

The estimates are nearly identical to the original model, showing that accounting for the site really didn't improve on the baseline model in a meaningful way. I also looked at the CIs for both models, which overlap, showing that there is no statistical difference.

Once that was done, I moved on to the final analysis. I had found the variables that are most associated with CAD, but now I want to know how well the model can predict CAD in a patient that the model hasn't encountered before. To do this, I selected the first of the 20 imputed datasets I had created before. I divided the 920 patients into 5 groups stratified by target, which ensures that all 5 groups have roughly the same proportion of CAD patients. I chose to work with 5 folds, so for each fold, a different group would be the test group, while the other 4 would be the train group. Essentially each group gets to be the test group once. Each fold fits 2 models on the training data: a mixed model (with the site intercept) and a fixed model (without the site intercept). Then both models get tested on the test group of that fold, which will output two AUC values. In total there will be 10 AUC values, 5 for the mixed model and 5 for the fixed model, after all 5 folds are over. The 5 AUC values per model were then averaged, resulting in:

Mixed AUC: 0.921 ± 0.02

Fixed AUC: 0.919 ± 0.022

This means that both models are great at predicting CAD, and further iterates that accounting for sites doesn't improve the base model. To prove this, I ran a paired t test (since the folds of the mixed and fixed models use the same training and testing data) which gave a p-value of 0.058, which is not significant.

References:

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